Chapter: Test of Hypothesis

Question No. 1

a) A soft drink company claims that the average amount of drink in its bottle is 500 ml. A random sample of 36 bottles shows an average amount of 492 ml. The population standard deviation is known to be 24 ml.

Requirement: At the 0.05 significance level, test whether the average amount of drink is less than 500 ml.

Given:

- Claimed mean: $\mu_0 = 500$ ml
- Sample size: $n = 36$
- Sample mean: $\bar{x} = 492$ ml
- Population standard deviation: $\sigma = 24$ ml
- Significance level: $\alpha = 0.05$
- Test type: left-tailed (average less than 500)

Hypotheses:

$$H_0: \mu = 500$$
$$H_1: \mu < 500$$

Test statistic (z-test):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{492 - 500}{24/\sqrt{36}} = \frac{-8}{24/6} = \frac{-8}{4} = -2$$

Critical value:
For $\alpha = 0.05$ (left-tailed), $z_{\text{crit}} = -1.645$

Decision:
Since $z = -2 < -1.645$, we reject H_0.

Conclusion:
There is sufficient evidence at the 0.05 significance level that the average amount of drink is less than 500 ml.

b) A bank wants to compare the average monthly savings of customers using mobile banking and customers using traditional banking. A sample of 45 mobile banking users shows an average monthly saving of BDT 8,200, with a population standard deviation of BDT 1,500. A sample of 50 traditional banking users shows an average monthly saving of BDT 7,600, with a population standard deviation of BDT 1,800.

Requirement: At the 0.05 significance level, can we conclude that mobile banking users save more on average?

Given:

- Mobile banking: $n_1 = 45$, $\bar{x}_1 = 8200$ BDT, $\sigma_1 = 1500$ BDT
- Traditional banking: $n_2 = 50$, $\bar{x}_2 = 7600$ BDT, $\sigma_2 = 1800$ BDT
- Significance level: $\alpha = 0.05$
- Test type: right-tailed (mobile users save more)

Hypotheses:

$$H_0: \mu_1 = \mu_2$$
$$H_1: \mu_1 > \mu_2$$

Test statistic (two-sample z-test):

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{8200 - 7600}{\sqrt{\frac{1500^2}{45} + \frac{1800^2}{50}}} = \frac{600}{\sqrt{\frac{2{,}250{,}000}{45} + \frac{3{,}240{,}000}{50}}} = \frac{600}{\sqrt{50{,}000 + 64{,}800}}$$

$$= \frac{600}{\sqrt{114{,}800}}$$

$$\sqrt{114{,}800} \approx 338.82 \Rightarrow z \approx \frac{600}{338.82} \approx 1.771$$

Critical value:
For $\alpha = 0.05$ (right-tailed), $z_{\text{crit}} = 1.645$

Decision:
Since $z \approx 1.771 > 1.645$, we reject H_0.

Conclusion:
There is sufficient evidence at the 0.05 significance level that mobile banking users save more on average than traditional banking users.

Question No. 2

a) A restaurant claims that the average waiting time for customers is 15 minutes. A sample of 25 customers shows an average waiting time of 17 minutes, with a sample standard deviation of 4 minutes.

Requirement: At the 0.05 significance level, test whether the average waiting time is greater than 15 minutes.

Given:
$\mu_0 = 15$ minutes, $\bar{x} = 17$, $s = 4$, $n = 25$, $\alpha = 0.05$

Step 1 – Hypotheses
$H_0: \mu = 15$ vs $H_1: \mu > 15$ (right-tailed)

Step 2 – Test Statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{17 - 15}{4/\sqrt{25}} = \frac{2}{0.8} = 2.50$$

Step 3 – Critical Value
$df = n - 1 = 24$, $\alpha = 0.05$ (one-tailed) → $t_{0.05,24} = 1.711$

Step 4 – Decision
$t = 2.50 > 1.711$ → Reject H_0

Conclusion: At the 0.05 significance level, there is sufficient evidence that the average waiting time exceeds 15 minutes.

b) A training institute claims that students score an average of 75 marks after completing its preparation course. A sample of 40 students shows an average score of 72 marks. The population standard deviation is 8 marks.

Requirement: At the 0.05 level of significance, can we conclude that the average score is less than 75 marks?

Given:
$\mu_0 = 75$, $\bar{x} = 72$, $\sigma = 8$, $n = 40$, $\alpha = 0.05$

Step 1 – Hypotheses
$H_0: \mu = 75$ vs $H_1: \mu < 75$ (left-tailed)

Step 2 – Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{72 - 75}{8/\sqrt{40}} = \frac{-3}{1.2649} = -2.37$$

Step 3 – Critical Value
$\alpha = 0.05$ (left-tailed) → $z_\alpha = -1.645$

Step 4 – Decision
$z = -2.37 < -1.645$ → Reject H_0

Conclusion: At the 0.05 level, there is sufficient evidence that the average score is less than 75 marks.

Chapter: Sampling Distribution and Estimation

Question No. 3

a) A mobile operator is planning to launch a new internet package. A survey of 600 customers shows that 390 customers are interested in purchasing the package.

Requirement:

i. Estimate the population proportion of customers interested in the package.

ii. Construct a 95% confidence interval for the population proportion.

iii. Interpret the result.

Given: $n = 600$, $x = 390$ customers interested.

(i) Point Estimate

$$\hat{p} = \frac{x}{n} = \frac{390}{600} = 0.65\ (65\%)$$

(ii) 95% Confidence Interval

$$\hat{q} = 1 - \hat{p} = 0.35, SE = \sqrt{\frac{\hat{p}\hat{q}}{n}} = \sqrt{\frac{0.65 \times 0.35}{600}} = 0.0195$$

$$CI = \hat{p} \pm z_{0.025} \cdot SE = 0.65 \pm 1.96 \times 0.0195 = 0.65 \pm 0.0382$$

$$CI = (0.6118,\ 0.6882)\text{or}(61.18\%,\ 68.82\%)$$

(iii) **Interpretation**
We are 95% confident that the true proportion of customers interested in the new internet package lies between 61.18% and 68.82%.

b) The manager of a garment factory wants to estimate the average daily output per worker. A sample of 30 workers shows an average output of 48 pieces per day, with a sample standard deviation of 6 pieces.

Requirement:

i. What is the best estimate of the population mean?
ii. Which distribution should be used for constructing the confidence interval? Explain briefly.
iii. For a 95% confidence interval, identify the appropriate test statistic.
iv. Construct the 95% confidence interval for the population mean.
v. Would it be reasonable to conclude that the population mean is 50 pieces? What about 60 pieces?

Given: $\boldsymbol{n = 30}$, $\boldsymbol{\bar{x} = 48}$ pieces, $\boldsymbol{s = 6}$ pieces.

(i) Best Estimate
$\boldsymbol{\bar{x} = 48}$ pieces/day

(ii) Distribution
Since $\boldsymbol{\sigma}$ unknown and $\boldsymbol{n = 30}$, use t-distribution with $\boldsymbol{df = 29}$.

(iii) Critical Value
For 95% CI, $\boldsymbol{t_{0.025,29} = 2.045}$

(iv) Confidence Interval

$$\boldsymbol{SE = \frac{s}{\sqrt{n}} = \frac{6}{\sqrt{30}} = 1.0954, CI = 48 \pm 2.045 \times 1.0954 = 48 \pm 2.24}$$

$\boldsymbol{CI = (45.76,\ 50.24)}$ pieces/day

(v) **Interpretation**

50 pieces falls inside the interval → reasonable.

60 pieces is above the upper bound → not reasonable.

Question No. 4

a) A poultry farm owner wants to estimate the average monthly feed consumption per chicken. A sample of 20 chickens shows an average consumption of 4.5 kg, with a sample standard deviation of 0.8 kg.

Requirement:

i. What is the best estimate of the population mean?

ii. Which distribution should be used for this estimation problem?

iii. What assumption is required?

iv. Construct a 99% confidence interval for the population mean.

v. Would it be reasonable to conclude that the population mean is 4 kg? What about 6 kg?

Given: $\boldsymbol{n = 20}$, $\boldsymbol{\bar{x} = 4.5}$ kg, $\boldsymbol{s = 0.8}$ kg.

(i) Best Estimate
$\boldsymbol{\bar{x} = 4.5}$ kg

(ii) Distribution
t-distribution with $\boldsymbol{df = 19}$

(iii) Assumption
Monthly feed consumption is normally distributed in the population.

(iv) 99% Confidence Interval

$$\boldsymbol{t_{0.005,19} = 2.861}$$

$$\boldsymbol{SE = \frac{0.8}{\sqrt{20}} = 0.1789, CI = 4.5 \pm 2.861 \times 0.1789 = 4.5 \pm 0.5118}$$
$$\boldsymbol{CI = (3.9882,\ 5.0118)}\text{ kg}$$

(v) **Reasonableness**

4 kg is below the lower bound → not reasonable.

6 kg is above the upper bound → not reasonable.

b) A supermarket is considering introducing a new loyalty card. A random sample of 500 customers shows that 340 customers are willing to use the loyalty card.

Requirement:

i. Estimate the population proportion.

ii. Construct a 95% confidence interval for the population proportion.

iii. Interpret your findings in the context of the problem.4(a) 99% CI for Mean (Small Sample)

Given: $\boldsymbol{n} = \mathbf{500}$, $\boldsymbol{x} = \mathbf{340}$ willing.

(i) Point Estimate

$$\hat{\boldsymbol{p}} = \frac{\mathbf{340}}{\mathbf{500}} = \mathbf{0.68}\ (\mathbf{68}\%)$$

(ii) 95% Confidence Interval

$$\boldsymbol{SE} = \sqrt{\frac{\mathbf{0.68 \times 0.32}}{\mathbf{500}}} = \mathbf{0.0209}, \boldsymbol{CI} = \mathbf{0.68 \pm 1.96 \times 0.0209 = 0.68 \pm 0.0409}$$

$$\boldsymbol{CI} = (\mathbf{0.6391},\ \mathbf{0.7209})\text{or}(\mathbf{63.91}\%,\ \mathbf{72.09}\%)$$

(iii) **Interpretation**

We are 95% confident that between 63.91% and 72.09% of all customers would use the loyalty card. Since the entire interval lies above 50%, the card is commercially viable.

Chapter: Correlation Analysis

Question No. 5

a)

The correlation coefficient ranges from −1.00 to +1.00. Explain this range graphically and interpret positive, negative, and zero correlation.

b)

Two supervisors rated the performance of 8 employees using an ordinal scale: Excellent, Good, Average, Fair.

Employee	Supervisor A Rating	Supervisor B Rating
1	Excellent	Good
2	Good	Good
3	Average	Average
4	Excellent	Excellent
5	Fair	Average
6	Good	Fair
7	Average	Good
8	Fair	Fair

Requirement:

Calculate the Spearman rank correlation coefficient. Do the two supervisors agree in their ratings?

5(a) Range and Interpretation of Pearson's r

$$-1 \leq r \leq +1$$

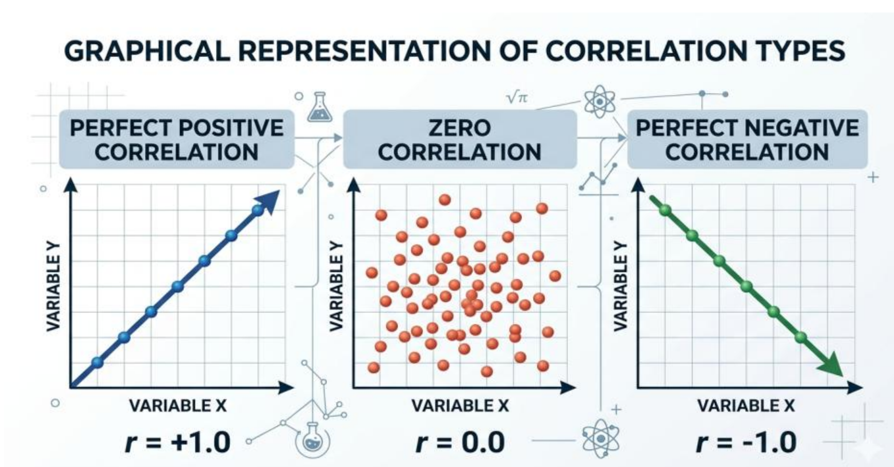
$r = +1$: perfect positive linear relationship

$0 < r < +1$: positive linear relationship

$r = 0$: no linear relationship

$-1 < r < 0$: negative linear relationship

$r = -1$: perfect negative linear relationship



5(b) Spearman Rank Correlation

Given:

8 employees rated by two supervisors (1 = Excellent, 2 = Good, 3 = Average, 4 = Fair).

Employee	Rating A	Rating B	Rank A	Rank B	d	d^2
1	Excellent	Good	1	2	-1	1
2	Good	Good	2	2	0	0
3	Average	Average	3	3	0	0
4	Excellent	Excellent	1	1	0	0
5	Fair	Average	4	3	1	1
6	Good	Fair	2	4	-2	4
7	Average	Good	3	2	1	1
8	Fair	Fair	4	4	0	0

$$\sum d^2 = 7, n = 8$$

$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 7}{8 \times 63} = 1 - \frac{42}{504} = 1 - 0.0833 = 0.9167$$

Conclusion: $r_s = 0.9167$ indicates a very strong positive agreement between the two supervisors.

Chapter: Regression Analysis

Question No. 6

The following data show annual sales and advertising expenditure of a company for 10 years.

Year	Sales (BDT Lakhs)	Advertising Expenditure (BDT Thousands)
2016	60	15
2017	75	20
2018	80	25
2019	90	30
2020	95	35
2021	105	40
2022	110	45
2023	125	50
2024	130	55
2025	140	60

Requirement:

a) Draw a scatter diagram and interpret the relationship between advertising expenditure and sales

b) Formulate and interpret a two-variable regression model to estimate the effect of advertising expenditure on sales. Also estimate sales if advertising expenditure is BDT 75 thousand.

Given data (10 years):

Year	Adv (X) BDT lakh	Sales (Y) BDT lakh
2016	15	60
2017	20	75
2018	25	80
2019	30	90
2020	35	95
2021	40	105
2022	45	110
2023	50	125
2024	55	130
2025	60	140

Calculations:

$$\sum X = 375, \sum Y = 1010, n = 10$$
$$\bar{X} = 37.5, \bar{Y} = 101$$
$$S_{xx} = \sum(X - \bar{X})^2 = 2062.5, S_{xy} = \sum(X - \bar{X})(Y - \bar{Y}) = 3500.0$$
$$b = \frac{S_{xy}}{S_{xx}} = \frac{3500}{2062.5} = 1.6970$$
$$a = \bar{Y} - b\bar{X} = 101 - 1.6970 \times 37.5 = 101 - 63.6375 = 37.3625$$

Regression Equation:

$$\hat{Y} = \mathbf{37.36 + 1.6970\,X}$$

Interpretation:

Slope $\boldsymbol{b} = \mathbf{1.6970}$: For every BDT 1 lakh increase in advertising, sales increase by approximately BDT 1.70 lakhs.

Intercept $\boldsymbol{a} = \mathbf{37.36}$: Estimated sales when advertising is zero (baseline).

Prediction for $\boldsymbol{X} = \mathbf{75}$:

$$\hat{Y} = \mathbf{37.36 + 1.6970 \times 75 = 37.36 + 127.275 = 164.635} \text{ BDT lakh}$$

Correlation $\boldsymbol{r} = \mathbf{0.9958}$ → very strong positive linear relationship.

Chapter: Probability

Question No. 7

a) A factory has three machines: A, B, and C. Machine A produce 30%, Machine B produces 45%, and Machine C produces 25% of the total output. The defective rates are 2%, 3%, and 5% respectively.

Requirement:

i. Draw a tree diagram.

ii. Find the probability that a randomly selected product is defective.

iii. If a product is defective, find the probability that it came from Machine C.

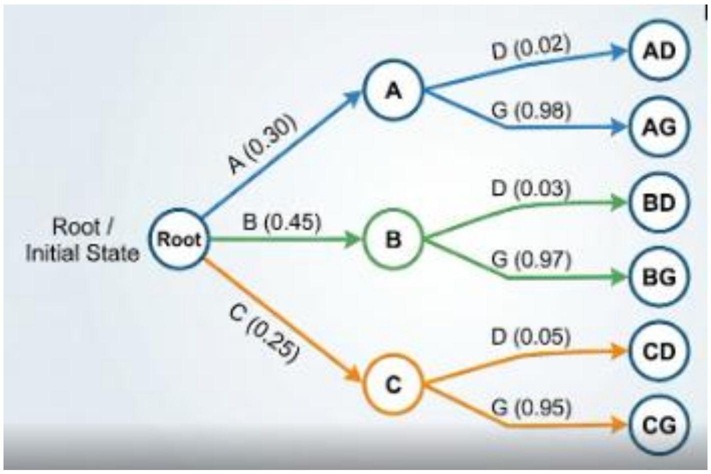
Given:

$$\boldsymbol{P(A) = 0.30,\ P(B) = 0.45,\ P(C) = 0.25}$$
$$\boldsymbol{P(D \mid A) = 0.02,}$$

$$\boldsymbol{P(D \mid B) = 0.03,}$$

$$\boldsymbol{P(D \mid C) = 0.05}$$

(i) Tree Diagram:



(ii) Total Probability of Defective

$$\boldsymbol{P(D) = 0.30 \times 0.02 + 0.45 \times 0.03 + 0.25 \times 0.05 = 0.006 + 0.0135 + 0.0125 = 0.0320}$$

(iii) Bayes – Probability Defective came from C

$$\boldsymbol{P(C \mid D) = \frac{P(C)P(D \mid C)}{P(D)} = \frac{0.0125}{0.0320} = 0.390625 \approx 39.1\%}$$

b) The weight of packets of rice produced by a company is normally distributed with a mean of 5 kg and a standard deviation of 0.12 kg. A packet is acceptable if its weight is between 4.80 kg and 5.20 kg.

Requirement:

Find the probability that a randomly selected packet is acceptable.7(a) Bayes' Theorem (Defective Products)

Given: $\boldsymbol{X \sim N(\mu = 5}$ kg, $\boldsymbol{\sigma = 0.12}$ kg), acceptable $\boldsymbol{4.80 \le X \le 5.20}$

Standardize:

$$\boldsymbol{z_1 = \frac{4.80 - 5.00}{0.12} = -1.6667 \approx -1.67, z_2 = \frac{5.20 - 5.00}{0.12} = 1.6667 \approx 1.67}$$
$$\boldsymbol{P(4.80 \le X \le 5.20) = P(-1.67 \le Z \le 1.67) = \Phi(1.67) - \Phi(-1.67)}$$

From z-table: $\boldsymbol{\Phi(1.67) = 0.9525,\ \Phi(-1.67) = 0.0475}$

$$P = 0.9525 - 0.0475 = 0.9050\ (90.50\%)$$

Interpretation: About 90.5% of packets are within the acceptable weight range.

Question No. 8

a) A manufacturer knows that 3% of its bulbs are defective. If an inspector randomly selects 12 bulbs, find the probability that at least one bulb is defective.

Given: $\boldsymbol{p = 0.03}$, $\boldsymbol{n = 12}$, let $\boldsymbol{X}$ = number defective.

$$\boldsymbol{P(X \geq 1) = 1 - P(X = 0) = 1 - (0.97)^{12} = 1 - 0.6938 = 0.3062}$$

≈ 30.6%

b) In a survey of 200 students, 70% use Facebook, 50% use Instagram, and 30% use both. If a student is selected randomly and it is known that the student uses Facebook, find the probability that the student also uses Instagram.

Given: $\boldsymbol{n = 200}$, $\boldsymbol{P(F) = 0.70}$, $\boldsymbol{P(I) = 0.50}$, $\boldsymbol{P(F \cap I) = 0.30}$

$$\boldsymbol{P(I \mid F) = \frac{P(F \cap I)}{P(F)} = \frac{0.30}{0.70} = 0.4286\ (42.86\%)}$$

If a student uses Facebook, there is a 42.86% probability they also use Instagram.

c) The lifetime of batteries produced by a company is normally distributed with a mean of 1,200 hours and a standard deviation of 80 hours. A battery is considered standard if its lifetime is between 1,100 hours and 1,350 hours.

Requirement:

Calculate the probability that a randomly selected battery is standard.

Given: $\boldsymbol{X \sim N(\mu = 1200}$ hr, $\boldsymbol{\sigma = 80}$ hr), standard $\boldsymbol{1100 \leq X \leq 1350}$

$$\boldsymbol{z_1 = \frac{1100 - 1200}{80} = -1.25, z_2 = \frac{1350 - 1200}{80} = 1.875}$$
$$\boldsymbol{P = P(-1.25 \leq Z \leq 1.875) = \Phi(1.875) - \Phi(-1.25)}$$

From table: $\boldsymbol{\Phi(1.875) \approx 0.9697}$, $\boldsymbol{\Phi(-1.25) = 0.1056}$

$$P = 0.9697 - 0.1056 = 0.8641\ (86.41\%)$$

About 86.4% of batteries meet the standard lifetime.

Chapter: Measures of Central Tendency and Measures of Dispersion

Question No. 9

a)

A company surveyed small retail shops to find the number of customers served per day. The frequency distribution is given below:

Number of Customers	Frequency
20 up to 30	4
30 up to 40	10
40 up to 50	18
50 up to 60	12
60 up to 70	6

Requirement:

Estimate the mean, standard deviation, and coefficient of variation.

Given: Customers per day (n=50)

Class	Midpoint m	Freq f	fm	$m-\bar{x}$	$(m-\bar{x})^2$	$f(m-\bar{x})^2$
20-30	25	4	100	-21.20	449.44	1797.76
30-40	35	10	350	-11.20	125.44	1254.40
40-50	45	18	810	-1.20	1.44	25.92
50-60	55	12	660	8.80	77.44	929.28
60-70	65	6	390	18.80	353.44	2120.64
Sum		50	2310			6128.00

$$\bar{x} = \frac{\sum fm}{n} = \frac{2310}{50} = 46.20$$

$$s^2 = \frac{\sum f(m-\bar{x})^2}{n} = \frac{6128}{50} = 122.56 \Rightarrow s = \sqrt{122.56} = 11.07$$

$$CV = \frac{s}{\bar{x}} \times 100 = \frac{11.07}{46.20} \times 100 = 23.96\%$$

Interpretation: Average customers = 46.20 per day, relative variability = 23.96%.

b)

The following frequency distribution shows the delivery time of online orders:

Delivery Time in Days	Frequency
1–3	5
3–5	9
5–7	16
7–9	12
9–11	6
11–13	2

Requirement:

Construct a box plot for the above data. Comment on the shape of the distribution.

Given frequency distribution (n=50): Cumulative frequencies: 5, 14, 30, 42, 48, 50.

$\boldsymbol{Q_1}$ (12.5th value) → class 3-5:

$$\boldsymbol{Q_1 = 3 + \frac{12.5 - 5}{9} \times 2 = 4.67}$$

$\boldsymbol{Q_2}$ (25th value) → class 5-7:

$$\boldsymbol{Q_2 = 5 + \frac{25 - 14}{16} \times 2 = 6.38}$$

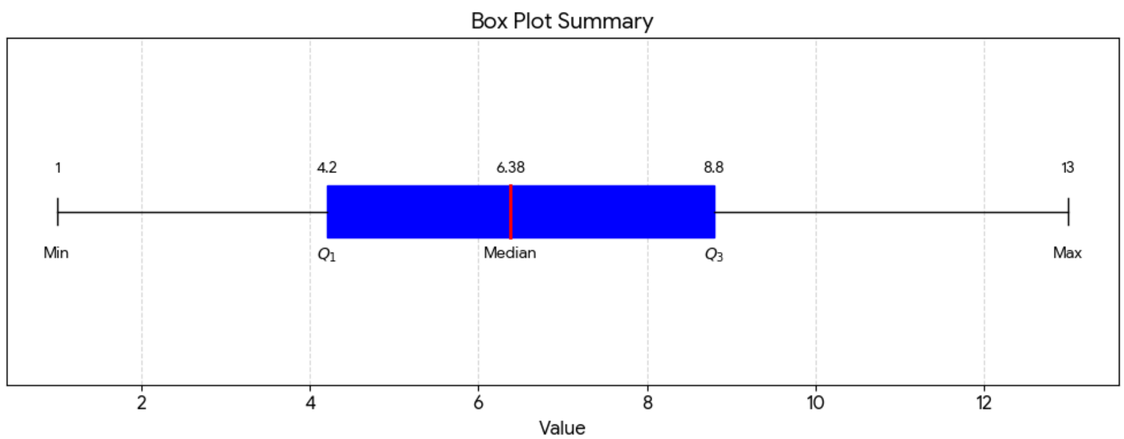
$\boldsymbol{Q_3}$ (37.5th value) → class 7-9:

$$\boldsymbol{Q_3 = 7 + \frac{37.5 - 30}{12} \times 2 = 8.25}$$

$$\boldsymbol{IQR = Q_3 - Q_1 = 8.25 - 4.67 = 3.58}$$

Lower fence = $\boldsymbol{Q_1 - 1.5 \times IQR = 4.67 - 5.38 = -0.71}$ → Min = 1

Upper fence = $\boldsymbol{Q_3 + 1.5 \times IQR = 8.25 + 5.38 = 13.62}$ → Max = 13



Question No. 10

a)

The following table shows the monthly expenditure of a sample of households:

Monthly Expenditure, BDT Thousand	No. of Households
10–20	8
20–30	16
30–40	30
40–50	24
50–60	14
60–70	8

Requirement:

Calculate the coefficient of skewness and interpret the value.

Given frequency distribution (n=100): (computations summarized)

$$\bar{x} = \mathbf{39.40}, \text{Median} = \mathbf{38.67}, \text{Mode} = \mathbf{37.00}, s = \mathbf{13.4402}$$

Pearson's coefficient of skewness:

$$Sk = \frac{3(\bar{x} - \text{Median})}{s} = \frac{3(39.40 - 38.67)}{13.4402} = \frac{2.19}{13.4402} = 0.1637 > 0$$

Interpretation: Distribution is positively skewed (right-skewed); tail extends to higher expenditures.

b)

The following frequency distribution gives the weights of packets produced by a machine:

Weight in Grams	Frequency
90–100	3
100–110	8
110–120	14
120–130	16
130–140	7
140–150	2

Requirement:

Construct a box plot and comment on the shape of the distribution.

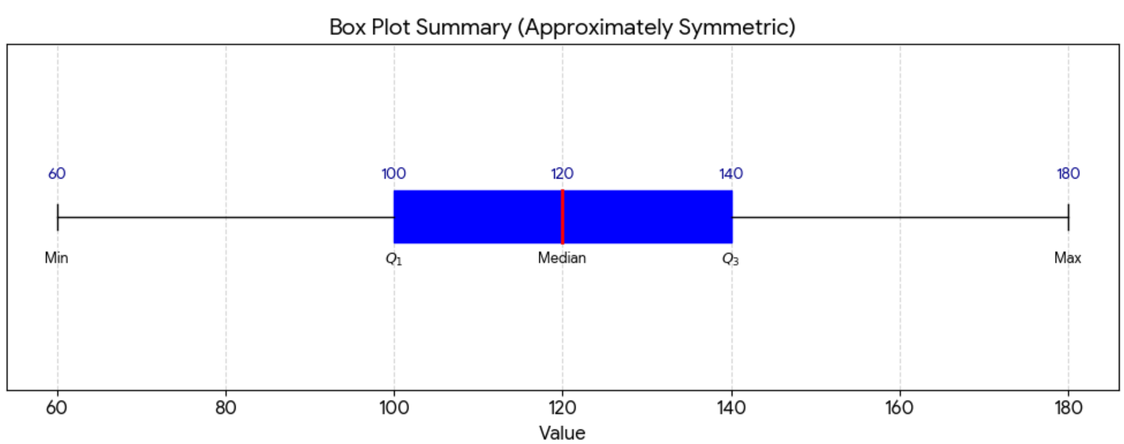
Cumulative frequencies: 3, 11, 25, 41, 48, 50 (n=50).

$$\boldsymbol{Q_1 = 100 + \frac{12.5-3}{8} \times 10 = 111.88} \text{ g}$$

$$\boldsymbol{Q_2 = 110 + \frac{25-11}{14} \times 10 = 120.00} \text{ g}$$

$$\boldsymbol{Q_3 = 120 + \frac{37.5-25}{16} \times 10 = 127.81} \text{ g}$$

$$\boldsymbol{IQR = 127.81 - 111.88 = 15.93}, \text{Min} = \boldsymbol{90}, \text{Max} = \boldsymbol{150}$$



Chapter: Introduction to Business Statistics, Collection of Data and Presentation of Data

Question No. 11

a)

A pie chart shows the types of transport used by 360 students to come to university. The percentages are as follows:

Transport Type	Percentage
Bus	40%
Rickshaw	25%
Bicycle	15%
Walking	20%

Requirement:

i. Calculate the number of students using buses.

ii. Calculate the number of students using rickshaws.

iii. Calculate the number of students who walk to university.

Transport	Percentage	Students (out of 360)
Bus	40%	144
Rickshaw	25%	90
Bicycle	15%	54
Walking	20%	72
Total	100%	360

b)

Identify the measurement scale: Nominal, Ordinal, Interval, or Ratio.

i. Blood group of patients

ii. Temperature in Celsius

iii. Monthly income of employees

iv. Ranking of students in a class

v. Brand names of mobile phones

vi. Number of children in a family

vii. Satisfaction level: Very satisfied, Satisfied, Neutral, Dissatisfied

viii. Calendar years

ix. Weight of rice bags

x. National ID numbers

Variable	Scale
Blood group	Nominal
Temperature (Celsius)	Interval
Monthly income	Ratio
Ranking of students	Ordinal
Brand names of phones	Nominal
Number of children	Ratio
Satisfaction level	Ordinal
Calendar years	Interval
Weight of rice bags	Ratio
National ID numbers	Nominal

c)
A cumulative frequency polygon shows the number of customers according to monthly purchase amount.
Requirement:
i. What is the class interval?
ii. What is the total number of customers studied?
iii. How many customers spent less than a given boundary value?
iv. How many customers spent between two given boundary values?

Construction: Plot upper class boundaries on X-axis vs. cumulative frequencies on Y-axis; connect with straight lines.

(i) Class interval: Difference between consecutive upper boundaries.

(ii) Total customers: Highest cumulative frequency (top-right endpoint).

(iii) Customers spending less than a boundary: Cumulative frequency at that boundary.

(iv) Customers between two boundaries: Difference of cumulative frequencies.

Question No. 12

a)

Write YES/NO depending on which measure of central tendency can be used for the following levels of measurement.

Level of Measurement Mode Median Mean

Nominal

Ordinal

Interval/Ratio

Level of Measurement	Mode	Median	Mean
Nominal	Yes	No	No
Ordinal	Yes	Yes	No
Interval/Ratio	Yes	Yes	Yes

b)

Identify the measurement scale: Nominal, Ordinal, Interval, or Ratio.

i. Customer feedback: Poor, Average, Good, Excellent

ii. Types of bank accounts

iii. Time shown on a digital clock

iv. Distance from home to workplace

v. Examination scores

vi. Shirt sizes: Small, Medium, Large

vii. Product serial numbers

viii. Age of respondents

ix. Political party preference

x. Year of birth

Variable	Scale
Customer feedback (Poor–Excellent)	Ordinal
Types of bank accounts	Nominal

Time on digital clock	Ratio
Distance home to workplace	Ratio
Examination scores	Interval/Ratio
Shirt sizes (S, M, L)	Ordinal
Product serial numbers	Nominal
Age of respondents	Ratio
Political party preference	Nominal
Year of birth	Interval

c)

The following table shows the number of hours students spent watching online lectures in one week:

Hours Spent	Frequency
0–2	4
2–4	10
4–6	20
6–8	12
8–10	4

Requirement:

Construct a histogram and draw a frequency polygon from it.

Data: Hours studied (n=50): 0-2(4), 2-4(10), 4-6(20), 6-8(12), 8-10(4).

Histogram: Adjacent rectangles of widths 2, heights equal to frequencies.

Frequency Polygon: Plot midpoints (1,3,5,7,9) vs frequencies (4,10,20,12,4); add anchor points (-1,0) and (11,0); connect with straight lines.

Short Questions

Question No. 13

Answer any ten of the following questions.

a) Define statistics.

b) What is the difference between primary data and secondary data?

c) What is meant by population and sample?

d) Define parameter and statistic.

e) What is a sampling distribution?

f) What is the standard error of the mean?

g) What is a confidence interval?

h) What is the meaning of 95% confidence level?

i) What is a null hypothesis?

j) What is an alternative hypothesis?

k) What is Type I error?

l) What is Type II error?

m) What is p-value?

n) What is level of significance?

o) What is meant by one-tailed test and two-tailed test?

(a) Statistics: Science of collecting, organising, analysing, and interpreting data.

(b) Primary vs Secondary: Primary = collected firsthand; Secondary = already exists.

(c) Population vs Sample: Population = entire group; Sample = subset.

(d) Parameter vs Statistic: Parameter describes population ($\boldsymbol{\mu}, \boldsymbol{\sigma}$); Statistic describes sample ($\bar{\boldsymbol{x}}, \boldsymbol{s}$).

(e) Sampling Distribution: Distribution of a statistic over all possible samples.

(f) Standard Error of Mean: $\boldsymbol{SE = \sigma/\sqrt{n}}$ – variability of sample means.

(g) Confidence Interval: Range likely containing the population parameter.

(h) 95% Confidence Level: 95% of such intervals contain the true parameter.

(i) Null Hypothesis ($\boldsymbol{H_0}$): Assumption of no effect or no difference.

(j) Alternative Hypothesis ($\boldsymbol{H_1}$): Statement contradicting $\boldsymbol{H_0}$.

(k) Type I Error: Rejecting true $\boldsymbol{H_0}$ (probability $\boldsymbol{\alpha}$).

(l) Type II Error: Failing to reject false $\boldsymbol{H_0}$ (probability $\boldsymbol{\beta}$).

(m) p-value: Probability of observed or more extreme result given $\boldsymbol{H_0}$ true.

(n) Level of Significance: Maximum probability of Type I error (usually 0.05).

(o) One-tailed vs Two-tailed: One-tailed tests direction; two-tailed tests any difference.

Question No. 14

Answer any ten of the following questions.

a) Define probability.

b) What is conditional probability?

c) State the addition rule of probability.

d) State the multiplication rule of probability.

e) What is Bayes' theorem?

f) What is a normal distribution?

g) Mention two properties of normal distribution.

h) What is z-score?

i) What is correlation?

j) What is the range of the correlation coefficient?

k) What does a negative correlation mean?

l) What is Spearman's rank correlation?

m) What is regression analysis?

n) What is the difference between dependent and independent variable?

o) What is a scatter diagram?

(a) Probability: Numerical measure of likelihood (0 to 1).

(b) Conditional Probability: $\boldsymbol{P(A \mid B) = P(A \cap B)/P(B)}$.

(c) Addition Rule: $\boldsymbol{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$.

(d) Multiplication Rule: $\boldsymbol{P(A \cap B) = P(A)P(B \mid A)}$.

(e) Bayes' Theorem: $\boldsymbol{P(A \mid B) = P(B \mid A)P(A)/P(B)}$.

(f) Normal Distribution: Symmetric, bell-shaped, defined by $\boldsymbol{\mu, \sigma}$.

(g) Properties: Symmetric, mean=median=mode, total area=1.

(h) Z-score: $\boldsymbol{Z = (X - \mu)/\sigma}$ – number of standard deviations from mean.

(i) Correlation: Strength & direction of linear relationship.

(j) Range of r: $\boldsymbol{-1 \leq r \leq +1}$.

(k) Negative correlation: One variable increases, the other decreases.

(l) Spearman's rank: $\boldsymbol{r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}}$ for ordinal data.

(m) Regression Analysis: Modelling relationship between dependent and independent variables.

(n) Dependent vs Independent: Dependent = outcome (Y); Independent = predictor (X).

(o) Scatter Diagram: Plot of (X,Y) pairs to visualise relationship.

Question No. 15

Answer any ten of the following questions.

a) Define mean, median, and mode.

b) What is range?

c) What is variance?

d) What is standard deviation?

e) What is coefficient of variation?

f) What is skewness?

g) What is a positively skewed distribution?

h) What is a negatively skewed distribution?

i) What is a box plot?

j) What are quartiles?

k) What is interquartile range?

l) What is a histogram?

m) What is a frequency polygon?

n) What is an ogive?

o) What is the difference between nominal and ordinal scale?

(a) Mean, Median, Mode: Mean = average; Median = middle value; Mode = most frequent.

(b) Range: Max $-$ Min.

(c) Variance: Average squared deviation from the mean.

(d) Standard Deviation: Square root of variance; original units.

(e) Coefficient of Variation: $\boldsymbol{CV = (s/\bar{x}) \times 100\%}$ – relative variability.

(f) Skewness: Asymmetry of distribution.

(g) Positive Skew: Tail right; mean > median > mode.

(h) Negative Skew: Tail left; mean < median < mode.

(i) Box Plot: Five-number summary: Min, $\boldsymbol{Q_1}$, Median, $\boldsymbol{Q_3}$, Max.

(j) Quartiles: Values dividing data into four equal parts (25th, 50th, 75th percentiles).

(k) Interquartile Range (IQR): $\boldsymbol{Q_3 - Q_1}$ – middle 50% of data.

(l) Histogram: Bar chart for continuous data, bars touch.

(m) Frequency Polygon: Line connecting midpoints of histogram bars.

(n) Ogive: Cumulative frequency polygon.

(o) Nominal vs Ordinal: Nominal = unordered categories; Ordinal = ordered categories.